

Performance Testing

Date	15 March 2026
Team ID	SWTID-2026-5650
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Performance Testing
Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust, Postman, k6
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	Comic Story Generation API & Gemini Model Integration Endpoint
Test Environment	Cloud / Hosted Server (GCP / AWS FastAPI Server)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal Load Testing on Comic Generation Endpoint	10	60	Response time < 2s, 0% error rate
2	Peak Load / Stress Testing on Gemini Story API	50	120	Response time < 5s, error rate < 1%
3	Spike Testing on Panel Image Rendering Engine	100	30	System recovers gracefully without crash
4	Endurance Testing for Multi-Panel Story Generation	20	300	Stable memory usage, no memory leaks

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.45 seconds	Pass	SLA met under normal load
2	Response Time (Max)	< 5 seconds	3.82 seconds	Pass	Handled peak stress effectively
3	Throughput (Req/sec)	> 15 req/sec	18.2 req/sec	Pass	Exceeds target processing rate
4	Error Rate	< 1%	0.2%	Pass	Minor transient timeout during spike
5	CPU Utilization	< 80%	64.5%	Pass	Optimal server load allocation
6	Memory Utilization	< 80%	58.1%	Pass	No memory leaks detected over 5m

Step 4: Observations & Analysis

Key Findings:

The ComicCraft AI Comic Story Creator API handled up to 50 concurrent virtual users gracefully with average response times (1.45s) well within the 2-second target SLA. Integration with Gemini models proved resilient under sustained request loads.

Bottlenecks Identified:

A momentary latency spike (up to 3.82s) occurred during the 100-user spike test specifically when rendering multi-panel high-resolution comic layouts concurrently.

Optimization Steps Taken:

- Implemented response caching for frequent Gemini prompt templates and character asset styles.
- Added asynchronous queueing for comic panel image rendering pipelines to prevent CPU saturation.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

